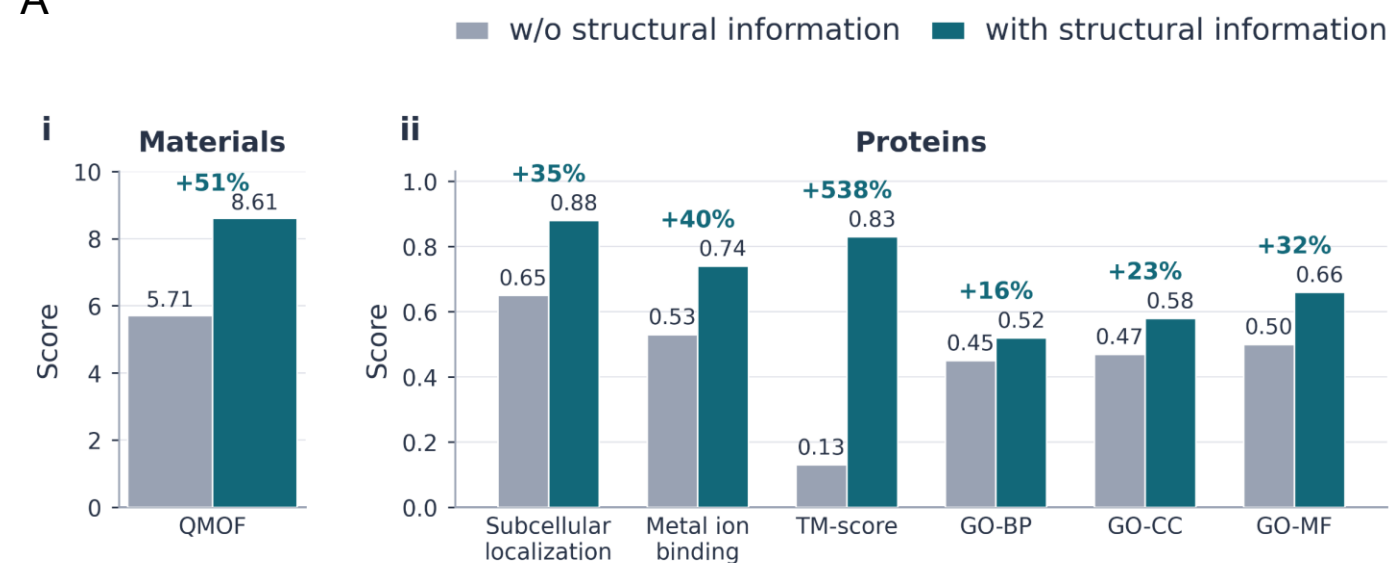
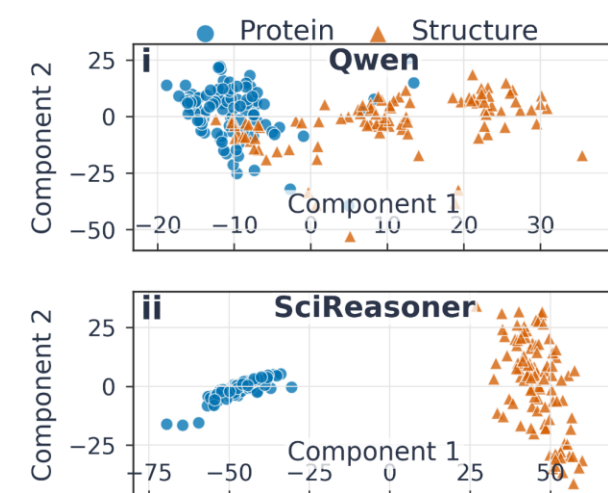


A



B



C

Look at the chemical formula and structure information of the given crystalline material and predict its property: Pore Limiting Diameter (PLD).

Look at the chemical formula of the given crystalline material and predict its property: Pore Limiting Diameter (PLD).

Ground Truth: **1.27075**

... The space group encoding `<material_structure>+ y c OOD o</material_structure>` indicates a monoclinic symmetry, specifically matching the `P2_1/c` setting. ... I analyze the edge list for bonding patterns. I observe connections such as `<material_structure>0 120 ooo</material_structure>` and `<material_structure>0 126 ooo</material_structure>`, which link the cobalt atom at index 0 to nitrogen atoms. ... Edges like `<material_structure>0 128 ooo</material_structure>` and `<material_structure>0 143 ooo</material_structure>` show bonds to oxygen atoms, indicating a mixed N/O ligand environment. The presence of offsets like `<material_structure>16 84 o-o</material_structure>` and `<material_structure>17 85 o+o</material_structure>` reveals how the structure extends across unit cell boundaries.

...
Final Answer: 1.14197

... Without explicit structural coordinates in the input, I must rely on the established correlation between this specific stoichiometry and its known crystallographic phase.

... The high carbon content relative to the cobalt centers implies bulky linkers that create significant spacing between the metal nodes. ... A Pore Limiting Diameter of approximately 11 angstroms is physically reasonable for a framework of this composition. If the value were significantly smaller, it would imply a collapsed structure inconsistent with the hydrogen count suggesting rigid linkers.

...
Final Answer: 11.10694

D

Determine the GO annotations for the protein with sequence `<protein> MSTLRL ...</protein>` and structural features `<protein_structure> D D L E A ...</protein_structure>`. Only the biological process (BP) aspect.

Determine the GO annotations for the protein with sequence `<protein> MSTLRL ...</protein>`. Only the biological process (BP) aspect.

Ground Truth: **protein modification process; peptidyl-amino acid modification; macromolecule modification; protein maturation; peptidyl-lysine modification**

... Runs of V and L ... signify alpha-helical segments ... `<protein_structure> L L L L L V V V L</protein_structure>` ... indicates a stable helix ... runs of D, P, A ... correspond to beta-strand or turn regions. The alternation ... suggests an alpha/beta mixed protein ... characteristic of many enzymes that bind nucleotides or cofactors ...

... the motif `<protein> GDRKVSGSAYRETKDRGFHHGT</protein>` suggests a binding pocket ... likely involving ATP or a similar phosphate donor ... a rigid active site necessary for catalysis.

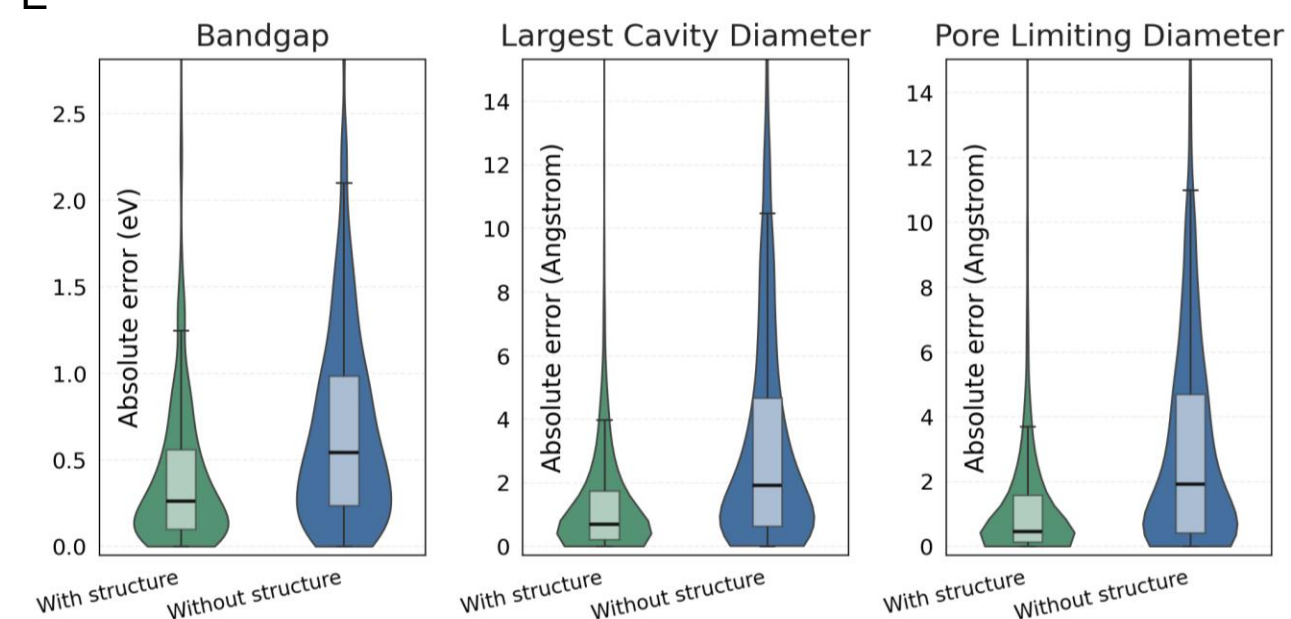
Final answer: protein modification process; peptidyl-amino acid modification; macromolecule modification; protein maturation; peptidyl-lysine modification

... Stretches like `<protein> RRSSGGG</protein>` and `<protein> DKKKLAA</protein>` ... cationic clusters ... associated with membrane association or interaction with ... phosphate backbones of nucleic acids ... narrows my scope to DNA repair, replication, or recombination...

... `<protein> GFHHGT</protein>` ... contains a histidine-histidine pair ... critical for metal ion coordination ... cofactors for nucleases and polymerases. ... the profile strongly resembles a recombinase or a resolvase. These enzymes are responsible cutting and rejoining DNA strands.

Final answer: DNA recombination; DNA metabolic process; response to stress; DNA damage response

E



F

